

# David Austin

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## EDUCATION

**Drexel University | Grad. 2023**  
*M.S. in Mechanical Engineering*  
*B.S. in Mechanical Engineering*  
3.91 GPA  
Pennoni Honors College  
Pi Tau Sigma Honors Society  
Executive Board | Drexel ASME

## CERTIFICATIONS

**Engineer-In-Training | NCEES**

## PATENTS

**Non-Invasive Measurement & Calculation System**  
U.S. Patent 12306048 | 2025

## CORE SKILLS

Heat Transfer Analysis  
Fluid & Piping Systems  
Component Design  
GD&T Application  
Structural/Stress Analysis  
Experimental Test Design

## SOFTWARE

**CAD:** SolidWorks, Fusion 360, AutoCAD, Creo, Inventor

**Analysis:** Ansys Mechanical, Fluent, MathCAD

**Productivity:** MS Office, Jira, Git, Linux/Unix Systems, LaTeX

**Coding:** Python (primary), VBA, R, MATLAB, Jupyter

## WORK EXPERIENCE

**Holtec International** **Camden, NJ**  
*Mechanical Design Engineer* *July 2023 – Present*

- Led development of 150+ drawing packages for equipment used in processing, transport, and storage of spent nuclear fuel
- Directed multi-disciplinary teams including thermal, structural, and manufacturing engineers to analyze and improve design proposals
- Served as technical lead for forced-convection drying systems, sealing/leak-testing equipment, custom high-temperature valves, vacuum systems, and other fluid-handling equipment
- Performed thermal-hydraulic modeling and developed R&D testing programs to qualify new equipment and processes

**Omega Engineering** **Bridgeport, NJ**  
*Mechanical Engineering Co-op* *November 2022 – June 2023*  
*March 2021 – March 2022*

- Designed and facilitated commercial launch of a patented non-invasive digital temperature meter for industrial piping systems
- Created and used prototypes and test fixtures for validation of new electro-mechanical designs and firmware for temperature meters
- Coordinated with product managers, technicians, software engineers, and material vendors to design new sensors and smart devices

**Cornelis Networks** **Chesterbrook, PA**  
*Mechanical Engineering Co-op* *March – September 2022*

- Conducted thermal and mechanical testing of computer server cooling systems including fans, heat pipes, and heat sinks
- Wrote Python software packages to automate server hardware tests and improve CAD data workflow in PTC Windchill

**Theoretical & Applied Mechanics Group** **Philadelphia, PA**  
*Advanced Manufacturing Researcher* *April 2019 – March 2021*

- Utilized FEA process simulations and topology optimization to counteract heat-related warping of 3D-printed titanium parts
- Simulated and prototyped customized 3D-printed biomedical devices which intentionally introduced warping to reduce fabrication time

## PUBLICATIONS

**Parametric Analysis of the Design Point for a Centrifugal Nuclear Thermal Rocket Fuel Element | American Nuclear Society | 2023**